After generating embeddings for all document chunks, the user's query is converted into an embedding using the same embedding model. Cosine similarity is then computed between the query embedding and the stored document embeddings to identify the most relevant chunks. These top-ranked chunks form the retrieval stage of the Retrieval-Augmented Generation (RAG) pipeline.

Once the relevant chunks are retrieved, they are combined with the user's query to construct a structured prompt, which is passed to a Large Language Model (LLM) for response generation. Initially, the **DeepSeek-R1** model was used for inference. However, due to its relatively slower response time on the available hardware, it was replaced with **Llama 3.2**, which provided significantly faster inference while maintaining high-quality, context-aware responses. This optimization improved the overall responsiveness and user experience of the RAG-based AI assistant without affecting retrieval accuracy.